

DSA STUDY BLUEPRINT SERIES

123. Number of Provinces Graph Problem

Identifying Unconnected Graph Components (Number of Provinces)

Section 10 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Problem:** Find count of disjoint subgraphs.
- Iterate vertices; run DFS/BFS starting at unvisited vertices, incrementing count.

Memory & Execution Blueprint



2 disconnected components (Provinces count = 2)

C++ Implementation Reference

Standard code layout and syntax

```
int findCircleNum(vector<vector<int>>& isConnected) {  
    int n = isConnected.size(), provinces = 0;  
    vector<bool> vis(n, false);  
    for (int i = 0; i < n; i++) {  
        if (!vis[i]) { provinces++; dfsProvinces(i, isConnected, vis); }  
    }  
    return provinces;  
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Time Complexity	$O(V^2)$ matrix scan
Space Complexity	$O(V)$

Key Practices & Gotchas

- **Important:** Each connected component represents an independent province.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace Dijkstra distance relaxation updates on active vertices:

Vertex popped	Adjacent edge checked	Current distance value	New path calculation	Relaxation action
U (Dist = 2)	U → V (Weight = 3)	V (Dist = 7)	New Dist = 2 + 3 = 5	Relax: Update V (Dist = 5)

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Shortest Path choices:** BFS finds unweighted short paths; Dijkstra handles positive weights; Bellman-Ford resolves negative edges.
- **Spanning Trees:** Prim's builds MST from a start node; Kruskal's sorts all edges using disjoint sets.
- **Related LeetCode Problems:** #200 Number of Islands, #743 Network Delay Time, #787 Cheapest Flights Within K Stops.